UNIT - II

NATURAL RESOURCES

Resource: Any material which can be transformed in a way that it becomes more valuable and useful.
'Any component of the natural environment that can be utilized by man to promote his welfare is considered as a natural resource'.
Land, Water, Minerals, Forests, Wildlife, Energy as well as Human beings are resources.

Classification of natural resources:

The quantity and quality of natural resources vary greatly in their location. Depending upon the availability and abundance, natural resources are of two types: inexhaustible and exhaustible.
1. Inexhaustible resources (Renewable resources): These resources are present in unlimited quantity in the nature and they are not likely to be exhausted by human activities. These are solar energy, wind power, rainfall, power of tides, hydro power, biological species.
2. Exhaustible resources (Non- renewable resources): These resources have limited supply on the earth, and are therefore, liable to be exhausted if used indiscriminately. Ex. Minerals, fossil fuels etc.

On the basis of origin, they are classified into two categories:
1. Biotic or organic/living resources: They are obtained from biosphere for example forest and forest products, crops, birds, animals, fish and other marine forms, coal, minerals, etc.
2. Abiotic or inorganic/non-living resources: Resources which are composed of non-living inorganic matter are called abiotic resources. Ex: land, water etc.

2.1.WATER RESOURCES

Rain is the main source of water on the earth. Clouds released water is called as precipitation. This is basically 3 forms – vaporized (drizzle), liquid (rain), solid form (snow). Water resources can be classified into surface water resources and ground water resources.
At global scale, about 71% of earth surface is covered with water. Total volume of water in hydrosphere is estimated to be 1.4 billion km^3 of which 97% is ocean water and rest 3 % is available as fresh water.
Even for fresh water availability we have to look for ice caps and glaciers as about 77.2% of fresh water is stored in them and remaining 22.4% is ground water and just 0.4% is disturbed in lakes, rivers & streams.
Main water resources in Andhra Pradesh are Godavari, Krishna and Pennar.

2.1.1. Water resources of the country:

Average annual rainfall	–	3000 billion m3
Average annual natural flows	-	1853 billion m3
Utilizable flows	-	1110 billion m3
Present utilization	-	176.13 billion m3

Ground water:

Utilizable groundwater	-	418 billion m3
Present utilization	-	100 billion m3

Total water resources (surface and ground water):

Utilizable water resources	-	1528 billion m3
Present utilization	-	276 billion m3

River flows in India:

Major rivers (>20,000 km^2 catchment's)	-	12
Medium rivers (2000 – 20,000 km^2 catchment's)	-	46
Minor rivers (<2,000 Km^2 catchment's)	-	innumerable.

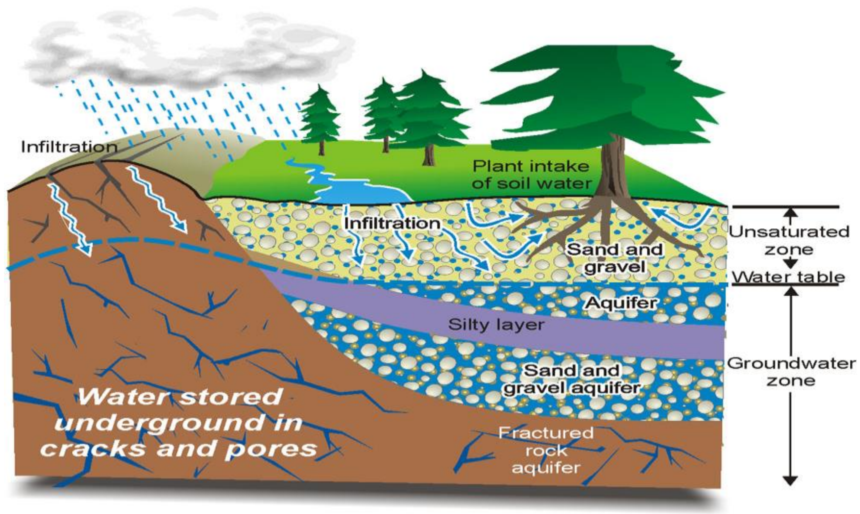
2.1.2.Forms of fresh water:

Fresh water mainly occurs in two forms: Ground water and Surface water.

1. Ground water: Ground water constitutes about 22.4% of the total fresh water resource. It is about 35-50 times that of surface water supplies. The ground water is contained in aquifers. An aquifer is a highly permeable layer of sediment or rock containing water. Layers of sand and gravel are good aquifers, while the clay and crystalline rocks (eg. Granite) having poor permeability are not good aquifers. Aquifers are of two types:

i) Un-confined aquifers: These are covered by permeable earth materials and are recharged by seeping down of water from rainfall and snowmelt.

ii) Confined aquifers: These are present between two impermeable layers of rock and are recharged only in those areas where the aquifer meets the land surface. The recharging area of many confined aquifers may be several kilometer away from the location of the well.



Under ground water system

2. Surface water: Surface water occurs in the forms of streams, rivers, lakes, ponds, wetlands and artificial reservoirs. It comes through precipitation (rain fall, snow) and accumulated in the water bodies, if not percolated into the ground or not returned to the atmosphere as evaporation or transpiration loss. Surface water mainly used for irrigation of crops, public water supply and industrial supply.

<u>Uses of surface and Ground water:</u>

Water is used mainly for two types of uses.

1. Consumptive use: Here water is completely utilized and it is not reused. Examples: in domestic application, industry and irrigation.
2. Non-consumptive use: Here water is not completely utilized and it is not reused. Examples: Hydropower.
3. Other important uses of water:

i. Water is mainly used for domestic purposes like drinking, cooking, bathing and washing etc.

ii. Water is also used for commercial purposes like hotels, theaters, educational institutions, offices etc.

iii. Another important use of water is for irrigation, like agriculture. Almost 60-70% of the fresh water is used for irrigation.

iv. 20 – 30% of the total fresh water is used for so many industrial operations like refineries, iron and steel, paper and pulp industries.

v. Water is very essential for the sustenance of all the living organisms.

vi. Water also plays a key role in sculpting the earth's surface, moderating climate and diluting pollutants.

<u>Over- utilization of surface and ground water:</u>

The rapid increase in population and industrial growth has increased demand, for water resources. Due to increase of ground water usage, the annual extraction of ground water is in far excess than the natural recharge.

Effects of over use of ground water:

a. Lowering of water table: Excessive use of ground water for drinking, irrigation and domestic purposes has resulted in rapid depletion of ground water in various regions leading to lowering of water table and drying of wells. The lowering of water table would cause a sharp decline in future agricultural production.

b. Ground subsidence: When ground water withdrawal is more than its recharge rate, the sediments in the aquifer become compacted. It is called ground subsidence. It results in sinking of overlying land surface which may damage buildings, destroy water supply, and reverse the flow of sewers and canals and tidal flooding.

c. Water logging: When excessive irrigation is done with brackish water it raises the water table gradually leading to water-logging and salinity problems.

d. Intrusion of salt water: In coastal area, over-exploitation of ground water would lead to rapid intrusion of salt water from sea.

e. Landslides and earthquakes: Over utilization of groundwater leads to decrease in water level, which cause earthquake, landslides and famine.

f. Pollution of water: When the ground water level near the agriculture land decreases, the, containing the nitrogen as nitrate fertilizer, percolates rapidly into the ground and pollute the ground water.

Problem: Water becomes unsuitable for potable use by infants, when nitrate concentration exceeds 45mg/l. It cause Methamoglobinemea (Blue baby) disease in less than one year babies.

2.1.3. Floods:

Inundation of a large land area with water for several days in continuation is called floods. Floods are caused by both natural as well as anthropogenic factors. Among natural factors, the important are prolonged downpour, blocking of free flow of the rivers due to siltation and landslides. The anthropogenic activities such as deforestation, over grazing, construction activities, channel manipulation through diversion of river course etc. are contributing largely to the sharp rise in the incidence of floods. Floods commonly occur when water from heavy rainfall, melting ice or snow or a combination of these, exceeds the capacity of the receiving river.

Melting of ice caps in polls is mainly due to increase earth temperature. This water flows into down areas. It submerges villages and agricultural fields. During the floods, the river carries fertile sediment and deposits it on the level land along its lower course. Such areas are called as flood plains. This is very fertile.

Flood control structures are not succeeding in India; only one or two structures are there. Reservoirs are storing the flood water. There is no dam to flood water.

Causes of floods:

1. Heavy rainfall, melting of snow, sudden release of water from dams, often causes floods in the low-lying coastal areas.
2. Prolonged downpour can also cause the over-flowing of lake and rivers resulting into floods.
3. Reduction in carrying capacity of the channel, due to accumulation of sediments or obstructions built on flood ways.

4. Deforestation, overgrazing, mining increases the runoff from rains and hence the level of flood raises.
5. The removal of dense and uniform forest cover over the hilly zones leads to occurrence of floods.

Effects of floods:

1. Due to flood, water spreads in the surrounding areas and submerges them.
2. Due to floods the plain surface have become eroded and silted with mud and sand, thus the cultivable land areas get affected.
3. Extinction of civilization in some coastal areas also occur.

Flood Management:

1. Floods an be controlled by constructing dams or reservoirs.
2. Channel management and embankments also control the floods.
3. Encroachment of flood ways should be banned.
4. Flood hazard may also be reduced by forecasting or flood warning.
5. Flood may also be reduced of runoff by increasing infiltration through appropriate afforestation in the catchment area.

2.1.4. Droughts:

Drought is lack or insufficiency of rain for an extended period, that causes considerable hydrologic imbalance and consequently water shortages, crop damage, stream flow reduction and depletion of groundwater levels and soil moisture.
Factors that interact to produce droughts:
Temperature, wind velocity, sunshine, soil texture, soil moisture. Main parameter is rainfall.
Human activities like deforestation, overgrazing, mining etc. are largely responsible for spreading of deserts, thereby converting more areas to drought affected areas.
The problems of drought can be mitigated by afforestation programmes, which increase the content of air moisture, the amount of precipitation and the rate of rain water infiltration.

National commission on agriculture (1976) classified the droughts into three types:

1. Meteorological drought:
   Situation when there is significant (more than 75%) decrease from normal precipitation over an area.
2. Hydrological drought:
   Meteorological drought, when prolonged, results in hydrological drought-marked depletion of surface water, drying up of reservoirs, lakes, streams/ rivers, fall of ground water levels.
3. Agricultural drought:
   Soil moisture and rainfall are inadequate for crop growth. Cause extreme crop stress and wilt.

Causes of drought:

1. When annual rainfall is below normal and less than evaporation, drought is created.

2. High population is also another cause for drought. Population growth leads to poor land use and makes the situation worse.

3. Intensive cropping pattern and over exploitation of scare water resources through dug well or bore well to get high productivity has converted drought prone areas into desert.

Ex. In Maharastra there has been no recovery from drought for the last 30 years due to over-exploitation of water by sugarcane crop.

4. Deforestation leads to desertification and drought too. When the trees are cut, the soil is subject to erosion by heavy rains, winds and sun. The removal of thin top layer of soil takes away the nutrients and the soil becomes useless. The eroded soils exhibit droughty tendency.

Effects of drought:

1. Drought causes hunger, malnutrition and scarcity of drinking water and also changes the quality of water.
2. Drought causes wider spread failures leading to acute shortage of food and adversely affects human and livestock populations.
3. The drought indicates the worst situation and initiation of desertification.
4. Raw materials for agro-based industries are critically affected during drought time, hence retarding the industrial and commercial growth.
5. Drought also accelerates degradation of natural resources.
6. Drought leads to large migration of people and urbanization.

Drought Management:

1. Indigenous knowledge in control of drought and desertification is very useful for dealing with the problems.
2. Rain water harvesting programme is another fruitful method to conserve more water and to control drought.
3. To improve ground water level, construction of reservoirs are essential in drought area.
4. Modern irrigation technology (drip irrigation) is very much useful to conserve water.
5. A forestation activity also improves the potential of water in the drought area.
6. Mixed cropping and dry farming are the suitable methods which minimize the risks of crop failures in dry area.

2.1.5. Dams:

Dams are built across the river in order to store water for irrigation, hydroelectric power generation and flood control. Most of the dams are built to serve for more than one purpose called “multi purpose dams”. These dams are called as the temples of modern India by the country’s first Prime Minister, Jawaharlal Nehru.

Surface water utilization we construct projects like Nagarjuna Sagar. This project useful for production of power and also it effects on people and environment.
Based on the size dams are three types – Large, Medium, Small dams.
Based on strength dams are rigid dams and non-rigid dams.
Rigid dams are concrete and masonry dam. Non-rigid dams are earth and rock dams.
Excess water is released to outside of the dam with out effect to dam. This is known as flood disposal, this water is called as surplus water.

<u>Problems/disadvantages of water resources projects:</u>
1. Submergence of forests with rare varieties of trees/plants.
2. Adverse effect on wild life and fish life habitat of rare species.
3. Effect on human health and sanitation.
4. Degradation in the fertility of the land – transport of sediment and fertilizer residual.
5. Rehabilitation of displaced persons.
6. Floods.
7. Some dams lose so much water through evaporation and seepage into porous rock beds that they waste more water than they make available.
8. Salts left behind by evaporation increase the salinity of the river and make its unusable when it reaches the down stream cities.
9. Growth of snail populations in the shallow permanent canals that distribute water to fields may lead to an epidemic of Schistosomiasis.

<u>Purposes of water resources projects:</u>
1. Irrigation during dry periods.
2. Ensuring a year-round water supply.
3. Hydroelectricity generation
4. Transfer of water from areas of excess to areas of deficit using canals.
5. Flood control and soil protection.
6. Drinking water.
7. Industrial needs,
8. Fish culture,
9. Recreation, Navigation, Tourism, etc.,

2.1.6. Case Studies:

<u>1. Tehri Dam:</u>
The 260.5 m high with 3.5 cu.km storage capacity Tehri Dam in Uttar Pradesh in India is being built at a cost of U$ 2 billion on narrow Himalayan gorge at the end of a wide trough shaped valley. Due to earth quake an October 20,1991, Uttar Kashi region close to the dam. Tehri Dam burst; it will create damage of un imaginable magnitude. Rishkesh will be under water in 63 minutes. Haridwar in 83 minutes, Bijnor in 4 hr and 45 minutes, Meerut in 7 hr and 25 minutes, etc.

2. Narmada Project:

This Rs. 25,000 crore world's largest river valley project consists of 30 major, 135 medium and 3000 minor dams. On the Narmada river and tributaries, flowing through M.P. Maharastra, Gujarat. Of the 30 major dams, 10 will be on Narmada and 20 on tributaries. The project will submerge 60,000 hectares of fertile land and rich forests. 11,300 hectares of agricultural land will also be submerged.66, 675 people (a majority tribal) have to be displaced.

The environmental damages of this project are – the displaced people will have to be rehabilitated in the forest area leading to loss of more forests. The out sees by and large have to migrate to the nearest cities in search of livelihood. This creates pressure on the urban life.

The compensatory measures are subjected to long delays and corrupt practices, denying justice to the victims of displacement.

On these grounds, the NGO's like Narmada Bachavo Andolan led by Medha Patkar have been fighting for the rights of the tribal displaced by the project. The controversy is still going on.

3. Sardar Sarovar project:

The Dam is situated on river Narmada and is spread over three states of Gujarat, Maharastra and M.P. It is one of the costiest project affecting villages in three states of M.P., Maharastra and Gujarat. About 245 villages will be submerged, of which about 193 are in M.P. alone. Over 75,000 (nearly 50,000 in M.P. alone) people will be evicted. Nearly 43,000 ha. of land will be needed for rehabilitation of the our sees in the Sardar Sarovar Project. A total of 1,44,731 ha of land will be submerged by the dam, out of which 56,547 ha is forest land.

2.2. MINERAL RESOURCES

2.2.1. Minerals:

A mineral is a naturally occurring inorganic crystalline solid. Minerals can be metallic e.g. iron, copper, gold etc. or non-metallic e.g. sand, stone, salt, phosphates etc. There are thousands of natural minerals like quartz, feldspar, calcite, topaz, bauxite composed of elements like oxygen, silicon, iron, Mg, aluminium, calcium etc. The eight most common minerals in the world are:

Iron : 33.3%, Oxygen : 29.8%, Silicon : 15.6%, Magnesium : 13.9%, Nickel : 2%, Calcium : 1.8%, Aluminium : 1.5%, Sodium : 0.2%.

Indian mineral resources:

India produces 64 minerals, including 4 fuel minerals, 11 metallic minerals and 49 non- metallic minerals.

Ex. Coal, petroleum, bauxite, iron, manganese etc.

2.2.2. Uses and exploitation:

Minerals find use in a large number of ways in every day use in domestic, agricultural, industrial and commercial sectors and thus form a very important part of any nation's economy.

Major reserves and important uses of some of the major metals:

Metal	Major world reserves	Major uses
Aluminium	Australia, Guinea, Jamaica	Packing food items, transportation, utensils, electronics.
Chromium	CIS, South Africa	For making high strength steel alloys, in textile/tanning industries.
Copper	U.S.A, Canada, CIS, Chili, Zambia	Electronic and electronic goods, building, construction, vessels
Iron	CIS, South America, Canada, U.S.A	Heavy machinery, steel production, transportation
Lead	North America, U.S.A., CIS.	Leaded gasoline, Car batteries, paints, ammunition.
Manganese	South Africa, CIS, Brazil,	For making high strength, high resistant steel alloys
Platinum	South Africa, CIS	Use in automobiles, catalytic converters, electronics, medical uses.
Gold	South Africa, CIS, Canada	Ornaments, medical use, electronic use, use in aerospace
Silver	Canada, South Africa, Mexico	Photography, electronics, jewelers
Nickel	CIS, Canada, New Caledonia	Chemical industry, steel alloys

Major uses of some non-metallic minerals:

Non-metal mineral	Major uses
Silicate minerals	Sand and gravel for construction, bricks, paving etc.
Lime stone	Used for concrete, building stone, used in agriculture for neutralizing acid soils, used in cement industry.
Gypsum	Used in plaster wall-board, in agriculture
Potash, Phosphorite	Used as fertilizers
Sulphur pyrites	Used in medicine, car battery, industry.

Benefits of minerals:

- Igneous rocks (igni means fire or agni) and non-metallic economic minerals like graphite, feldspar, quartz, diamond, that are valued for their beauty and rarity also are exploited badly.

Maximum resources are produced in South America, South Africa. But maximum minerals are consumed by USA, Japan, Europe and then others.

- Sand and gravel are extracted for brick and concrete construction, pavements, transportation, filling and sand blasting.
- Pure silica is used for glass manufacturing.
- Limestone is mixed for concrete, road making, and building stones.

- Gypsum (calcium sulphate) is used for plastering.
- Potassium for fertilizers.
- Sulphur in the form of Pyrite (FeS_2) for making H_2SO_4.

Some of the major minerals in India:

a. Energy generating minerals:

Coal and lignite: West Bengal, Jharkhand, Orissa, M.P., A.P.

Uranium (Pitchblende or Uranite ore): Jharkhand, A.P. (Nellore, Nalgonda), Meghalaya, Rajasthan (Ajmer).

b. Other commercially used minerals:

Aluminium (Bauxite ore): Jharkhand, West Bengal, Maharastra, M.P., Tamilnadu.

Iron (Haematite and Magnatite ore): Jharkhand, Orissa, M.P., A.P., Taminadu, Karnataka, Maharastra and Goa.

Copper (Copper Pyrites): Rajasthan (Khetri), Bihar, Jharkhand, Karnataka, M.P., West Bengal, A.P., and Uttaranchal.

2.2.3. Environmental effects of extracting and using mineral resources:

The Environmental effects of extracting and using mineral resources depend on such factors as ore quality, mining procedures, local hydrological conditions, climate, rock types, size of operation, topography and several other related factors.

Some of the major environmental impacts of mining and processing operations are:

- Degradation of land
- Pollution of surface and ground water resources due to the release of harmful trace elements (cadmium, cobalt, copper, lead, and others) by leaching, even if drainage is controlled.
- Serious adverse impact on the growth of vegetation due to leaching out of trace elements and minerals.
- Air pollution due to emission of dust and gases.
- Deforestation including loss of fauna and flora.
- Adverse impact on historical monuments and religious places.
- Physical changes in the land, soil, water and air associated with mining directly and indirectly affect the biological environment.
- Rehabilitation of affected population including tribal.

Dereliction (closing or abandoning mines, i.e. deserting and left to fall into ruin): The harmful effects of dereliction include:

- Waste of agricultural and industrial land and Ugliness
- Health and accident hazards: land over underground mines may subside, causing houses to collapse. And ground water pollution.
- Shafts that are not filled in may lead to accidents.
- Old quarries and open-cast pits may also be dangerous.
- Permanent damage to landscape.

Effects of minerals:

* Non metal minerals are mainly silicates i.e. sand, limestone and soils.
  Extracting and using of this minerals release the particulates into atmosphere.
  Particulates effect on human health and photosynthesis in plants.

* Metal minerals like Gypsum (Calcium sulphate), Potassium etc. are harmful to environment. Metals like Fe, Al, etc. are dangerous to crops.
* Natural deposits of fluoride such as topaz, fluorspar (CaF_2) and Criolite($Na_3Al F_6$) these contaminate the water.
* Fluoride (F) of up to 1mg/l is essential to prevent dental cavities in children.
  1.5 mg/l of fluoride causes yellowing of teeth called ‘motling’.
  >5 mg/l of fluoride causes fluorosis and skeletal abnormalities.

General stages of mineral exploitation with summaries of activities and environmental issues:

Development Phase	Potential activities	Environmental issues
1. Mine identification	Geochemical, geophysical surveys, drilling, trenching, road/helcapter transport, bulk sampling.	Habitat disruption, noise pollution, acid mine drainage etc.
2. Mining and milling	EIA, mine design and construction, crushing/ grinding of ore etc.	Wildlife and fisheries habitat loss, changes in local water balances wind borne dust etc.
3. Smelting and refining	Processing of mineral concentrate by heat or electro-chemical process.	SO2 emissions contribute to Acid rains etc.
4. Mine closure	Reencountering of pit walls and waste dumps, ongoing monitoring and possible water quality treatment.	Seepage of toxic solutions into ground and surface water contamination from Acid mine drainage, wild life and fisheries habitat loss etc.

2.2.4 Management of mineral resources:

1. The efficient use and production of mineral resources.
2. Modernization of the mining industries.
3. Search for new deposit.
4. Re-use and re-cycling of the metals.
5. Environmental impacts can be minimized by adopting eco-friendly mining technology.
6. The low-grade ores can be better utilized by using microbial leaching techniques.

2.2.5. Case Studies:

1. Mining and quarrying in Udaipur:

About 200 open cast mining and quarrying centers in Udaipur, about half of which are illegal are involved in stone mining including soapstone, building stone, rock phosphate and dolomite. The mines spread over 15,000 hectares in Udaipur have caused many adverse impacts on environment. About 150 tons of explosives are used per month in blasting. The over burden, wash off, discharge of mine water etc. pollute the water. The Maton mines have badly polluted the Ahar river. The hills around the mines are devoid of any vegetation except a few scattered patches and the hills are suffering from acute soil erosion. The waste water flows towards a big tank of 'Bag Dara'. Due to scarcity of water people are compelled to use this effluent for irrigation purpose. The blasting activity has adversely affected the fauna and the animals like tiger, lion, deer and even fox, wild cats and birds have disappeared from the mining area.

2. Mining in Sariska Tiger reserve in Aravallis:

The Aravalli range is spread over about 692 km in the NW India covering Gujarat, Rajasthan, Haryana and Delhi. The hill region is very rich in biodiversity as well as mineral resources. The Sariska tiger reserve has gentle sloppy hills, vertical rocky valley, flat plains as well as deep gorges. The reserve is very rich in wild life and has enormous mineral reserves like quartzite, Schists, marble and granite in abundance.

Mining operations within and around the Sariska Tiger reserve has left many areas permanently infertile and barren. The precious wild life is under serious threat. We must preserve the Aravalli series as a National Heritage and the Supreme Court on December 31st 1991 has given a judgment in response to a Public Interest Litigation of Tarun Bharat Sangh, an NGO wherein both Centre and State Government of Rajasthan have been directed to ensure that all mining activity within he park be stopped. More than 400 mines were shut immediately. But, still some illegal mining is in progress.

3. Uranium Mining in Nalgonda, A.P.- The public hearing: The present reserves of Uranium in Jaduguda mines, Jharkhand can supply the yellow cake only till 2004. There is a pressing need for mining more uranium to meet the demands of India's nuclear program. The Uranium Corporation of India (UCIL) proposes to mine uranium from the deposits in Lambapur and Peddagutta villages of Nalgonda district in A.P. and a processing unit at about 18 kms at Mallapur. He plan is to extract the ore of 11.02 million tons in 20 years. The proposed mines would cover about 445 ha of Yellapurum Reserve Forest and Rajiv Gandhi Tiger Sanctuary. The public hearing held just recently in February, 2004 witnessed strong protests from NGOs and many villagers. The fate of the proposed mining is yet to be decided.

2.3. ENERGY RESOURCES

2.3.1. Growing energy needs:

Energy is an important input for development. It aims at human welfare covering household, agriculture, transport and industrial complexes. The rapidly expanding human population, human energy needs are increasing rapidly. It will demand the exploitation of most energy sources. Like other natural resources, energy resources are also non-renewable as well as renewable.

India's per capita energy consumption today, at 350 KWH per annum, is one tenth of world's average and one fortieth of U.S. average. The energy consumption is concentrated in the urban sector covering 30% of the country's population.

For example- an electric bulb uses 100 watts of energy.

A color T.V. uses 200 watts,

A car uses about 30,000 watts.

Thousands of years ago people had only sun's energy and their own energy. They burnt wood for heat, used animals for pulling. Surely we'll be going to those days again, steadily and surely.

The domestic power consumption has the following pattern of percentages:

Purpose	Urban	Rural
Lighting by Kerosene	45.2	84.0
Lighting by electricity	53.0	--
Cooking by non commercial fuel	58.1	4.5
Cooking by Kerosene	26.5	--

The present electric power generating capacity of 1,00,000 MW if facing a 20% deficit to meet the demand.

The world consumes energy in the following fashion.

Petroleum products = 39%, Natural gas = 24%, Coal = 32%, Tidal Power = 2.5%, Nuclear power = 2.5%.

2.3.2. Classification of energy resources

Based on continual utility, energy resources can be classified into two types:

1. Non- renewable energy resources:

These are exhaustible resources and available in limited amount and develop over a longer period. These include-

a. **Fossil fuels:** like coal, petroleum, natural gas and mineral oil. Coal, petroleum and natural gas, the common sources of energy, being organic (biotic) in their origin are also called 'fossil fuels'. Fossil fuels contribute 60% of the present power generation.

i) Coal: Coal is the first fossil fuel to be exploited on a large scale; so industrial revolution enabled coal to be mined at even greater depths.

Types of coal:

Peat: contains 5% of carbon, 5% of volatile matter and 90% moisture. It is having low calorific value. It is generally in dry condition.
Lignite: is the lowest form of coal. It is brown in color. Hence it is called as brown coal. Pure lignite is mined at Neyveli, Tamilnadu. It has 38% of carbon, 19% of volatile matter and 43% of moisture.
Bituminous coal: is also called soft coal. It contains 3% of water. It is highly flammable. It has 65% of carbon, and 32% of volatile matter.
Anthracite: is a hard and dense coal. This type of coal contains 96% of carbon, 1% of volatile matter and 3% of moisture. It has the highest energy content of all coals and is used for space heating and generating electricity.

ii) Oil or Petroleum: Petroleum is an inflammable liquid composed of hydrocarbons, which constitute the majority and the remainder is inorganic compounds like O_2, N_2, S and traces of organic-metallic compounds.
It is pumped to the surface as crude oil and refined to get desired products.

Distillation

Heated crude oil ------------------→ Fuel gas, Propane, Butane, liquefied gas
→ Gasoline
→ Kerosene
→ Furnace oil
→ Diesel
→ Lubricants
→ Asphalt

Petroleum fractionation process

As a source of energy petroleum has many advantages

- it is relatively cheap to extract and transport
- it requires little processing to produce desired products and
- it has relatively high net and useful energy yield.

However it has certain disadvantages also,

- produces environmental pollution
- oil spills, in ocean cause water pollution and is expensive to clean up.

iii) Petroleum gas: It is the mixture of three hydrocarbons such as butane, propane and ethane. The main constituent of petroleum gas is butane. These are in gaseous state, which is liquefied under pressure. So it is called as LPG.
A domestic cylinder contains 14kg of LPG. A strong smelling substance called ethyl mercaptane is added to LPG cylinder to help in the detection of gas leakage.

iv) Natural gas: It consists of methane with small quantities of ethane and propane. It is available in deep under crust. Natural gas is a by-product to petroleum mining. It meets about 24% of the world's energy requirement. In our country few natural gas fields have been discovered recently in Tripura, shore area of Bombay and in the Krishna, Godavari delta.

b. **Nuclear fuels :** around 2.8 % of the power generation at present is atomic power with 10,000 MW. Radio active materials like Uranium, Thorium, Radium etc. metals are used for production of energy by fission and fusion process.

Energy is generated from uranium for the peaceful work. 1kg of Uranium liberates an energy equivalent to 35000kg of coal. Nuclear energy is used for generation of electricity and used to running submarines, warships, space-crafts etc.

The first nuclear reactor was started in USA on Dec 2, 1942. Today almost all countries have atomic reactors for power generation. In France, 73% of the commercial energy is provided by atomic reactors.

There are several types of nuclear reactors, those are-

i) Light Water Reactor (LWR)
ii) Heavy Water Reactor (HWR)
iii) Gas Cooled Reactor (GCR)
iv) Boiling Water Reactor (BWR)
v) Pressurized Water Reactor (PWR)
vi) Liquid Metal Fast Breeder Reactor (LMFBR)

Nuclear energy can be produced by two types of reactions.

1. By nuclear fission
2. By nuclear fusion.

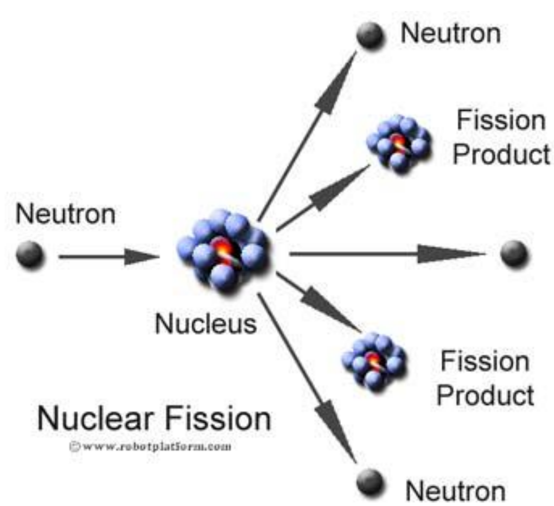
1. Nuclear fission: Nuclear fission is a nuclear chain in which heavier nucleus are split into lighter nuclei, on bombardment by fast moving neutrons and a large amount of energy is released through a chain reaction.

Ex: Fission of U^{235}

When U^{235} nucleus is hit by a thermal neutron, it undergoes the following reaction with the release of 3 neutrons.

$$ {}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{94}_{36}Kr + 3\,{}^{1}_{0}n + \text{Energy} $$

Each of the above 3 neutrons strikes another U^{235} nucleus causing (3 X 3) 9 subsequent reactions. These 9 reactions further give rise to (3 X 9) 27 reactions. This process of propagation of the reaction by multiplication in threes at each fission is called chain reaction.

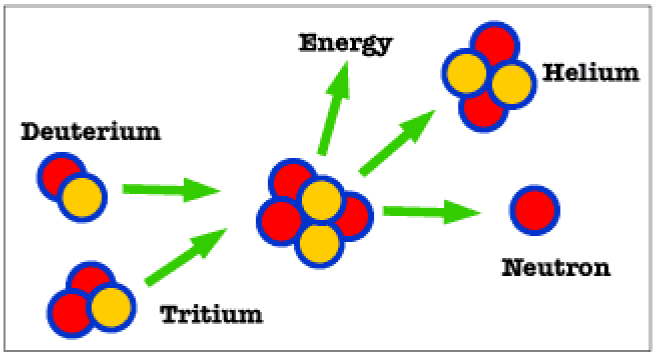


Schematic representation of chain reaction

2. Nuclear fusion: Nuclear fusion is a nuclear chain in which lighter nucleus are combined together at extremely high temperatures (1 billion ^{0}C) to form heavier nucleus and a large amount of energy is released.

Ex: Fusion of H^{2}_{1}

Two hydrogen-2 (Deuterium) atoms may fuse to form a helium at 1 billion ^{0}C with the release of large amount of energy.



2. Renewable energy resources:

These are mostly biomass based and available in unlimited amount in nature. These are inexhaustible materials.

These include:-

a. Hydro energy: Large amount of kinetic energy of the flowing water is tapped using water turbines. Hydro power projects involves construction of Dams. Currently 20% of the world's energy demand is produced in the form of hydro-energy. India is tapping about 84000MW through hydel potential. Totally, 271 projects have been installed in India. Advantages of Hydro power- clean source of energy, provides irrigation facilities, provides drinking water to the people living around.

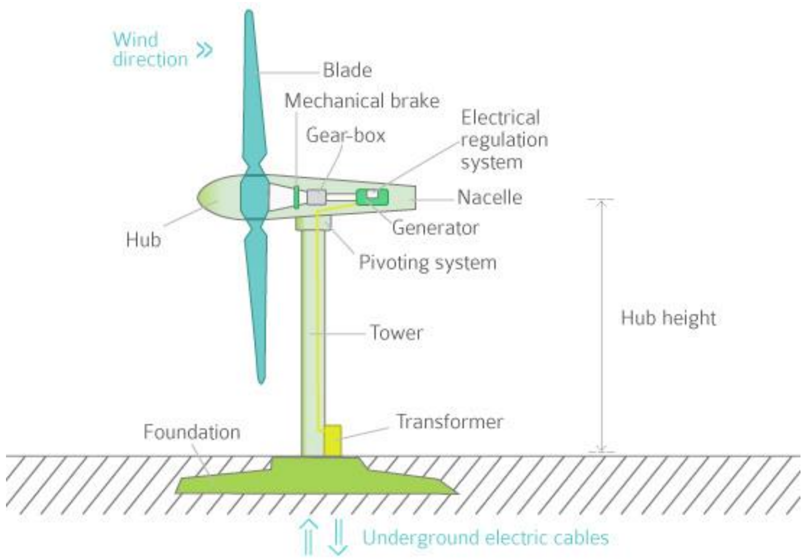
b. Wind energy : Wind forms in two major ways in the atmosphere.

- Unequal distribution of temperature, pressure on earth surface.
- Continuous rotation of earth.

Wind mills have a good potential as an energy source, particularly in rural areas. The mobilization of this resource at present is negligible. Most important advantages of wind energy is – a perennial sources, available all the day and night, eco-friendly, non-polluting and freely available.

The energy possessed by wind is because of its high speed. The wind energy is harnessed by making use of wing mills.

1. Wind mills: The strike of blowing wind on the blades of the wind mill makes it rotating continuously. The rotational motion of the blade drives a number of machines like water pump, flour mill and electric generators.

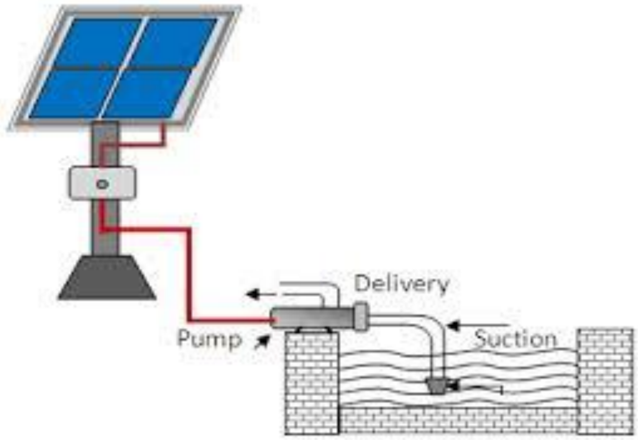


2. Wind Farms: When a large number of wind mills are installed and joined together in a definite pattern it forms a wind farm. The wind farms, produce a large amount of electricity.
Condition: The minimum speed required for satisfactory working of a wind generator is 15km/hr.
Advantages: It does not cause any air pollution, it is very cheap.

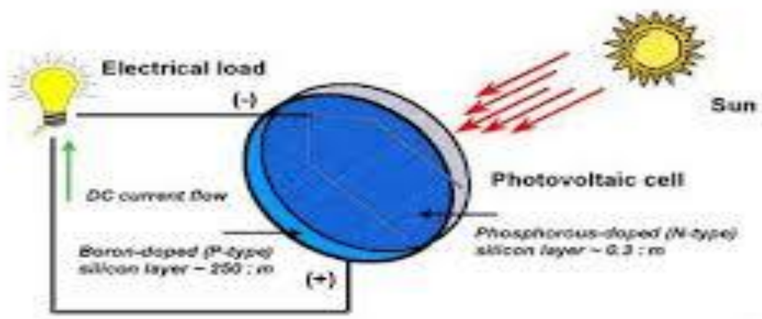
c. Solar energy: The sunshine in India supplies 1600 to 200 Kw per sq. m per year. Photovoltaic cell technology can hold a good promise for realizing the solar energy available in India. Presently it used to power satellites, watches, calculating devices etc. Following are some of applications of solar energy directly useful to the human beings- solar water heaters, solar cookers, solar drier, solar refrigerator.
The nuclear fusion reaction occurring inside the sun releases enormous amount of energy in the form of heat and light. Several techniques are available for collecting, converting and using solar energy.

Methods of harvesting solar energy:

i. Solar cells or Photovoltaic cells or PV cells: Solar cells consists of a p-type semiconductor (such as Si doped with B) and n-type semiconductor (such as Si doped with P). They are in close with each other. When the solar rays fall on the top layer of p-type semiconductor, the electrons from the valence band get promoted to the conduction band and cross the p-n junction into n-type semiconductor.



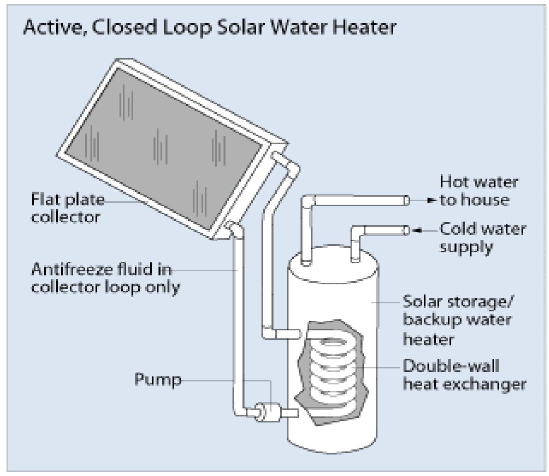
Solar pump



ii. Solar heat collectors: Solar heat collectors consist of natural materials like stones, bricks or materials like glass, which can absorb heat during the day time and release it slowly at night.

Uses: It is generally used in cold places, where houses are kept in hot condition using solar heat collectors.

iii. Solar water heater: It consists of an insulated box, inside of which is painted with black paint. It is also provided with a glass lid to receive and store solar heat. Inside the box it has black painted copper coil, through which cold water is allowed to flow in, which gets heated up and flows out into a storage tank. From the storage tank water is then supplied through pipes.



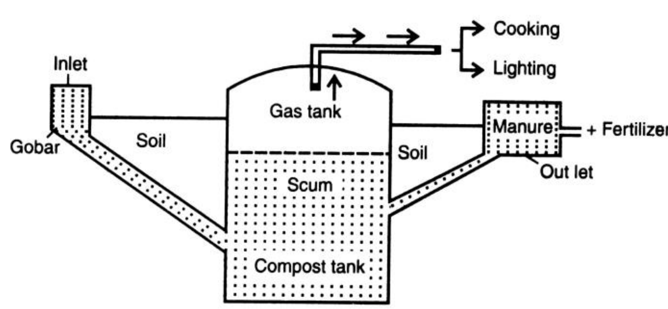
Solar water heater

d. Biomass: the matter originating directly or indirectly from photosynthesis is called biomass. Plant matter, bio degradable organic wastes, combustible waste residues etc constitute the biomass that can release heat energy on chemical/ biochemical oxidation.
Wood as fuel (amounting to 68.5%), oil products (15.6%), animal dung (8.3%) etc. are the biomass energy sources in the Indian rural area.

e. Bio gas: With 240 million cattle, India is the largest cattle dung producer. This material has a good potential to yield (methane) fuel gas under an anaerobic decomposition.
Bio gas producing potential of different live stock is- Cow dung - 100, Buffalo dung - 258, Camel dung - 283, Poultry droppings - 616

The composition of bio gas -	Methane	:	50 to 55%,
	Carbon dioxide	:	25 to 35%,
	Oxygen	:	6 to 8%
	Other gases	:	7 to 15%

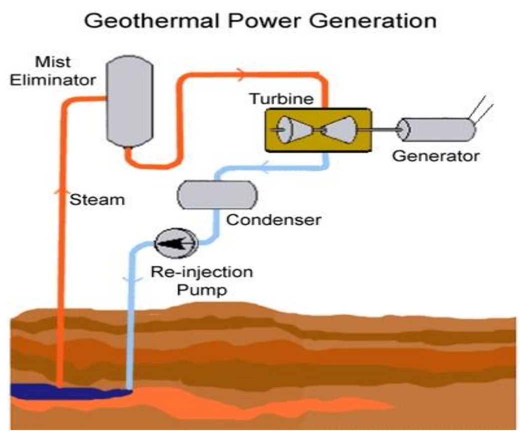
Over 1,50,000 bio gas plants are in operation today in our country.



Biogas Plant

f. Animal energy: Animal energy consumed in the agricultural sector of our country, employing 70 million bullocks, 8 million buffaloes, 1 million horses and 1 million camels in addition to donkeys, elephants. This animal power amounts to 30,000 MW as against 40,000 MW of total electrical energy capacity today.

g. Geothermal energy: The earth is having large amount geothermal energy with a higher temperature that is 4500deg.C. The energy comes from the molten rock material beneath the earth surface. The available thermal energy is used to produce power by running a smaller capacity thermal power plant.
The energy harnessed from the high temperature present inside the earth is called geothermal energy.



h. Tidal energy: Tides, the alternate rise and fall of sea water possess lot of energy. The identified tidal power potential in India is around 9000 MW. Currently Russia, China and Canada are effectively utilizing the tidal energy to produce 2 to 3% of their energy demand.
Ocean tides, produced by gravitational forces of sun and moon, contain enormous amount of energy. The 'high tide' and 'low tide' refer to the rise and fall of water in the oceans. The tidal energy can be harnessed by constructing a tidal barrage.
During high tide, the sea water is allowed to flow into the reservoir of the barrage and rotates the turbine, which in turn produces electricity by rotating the generators.
During low tide, when the sea level is low, the sea water stored in the barrage reservoir is allowed to flow into the sea and again rotates the turbine.

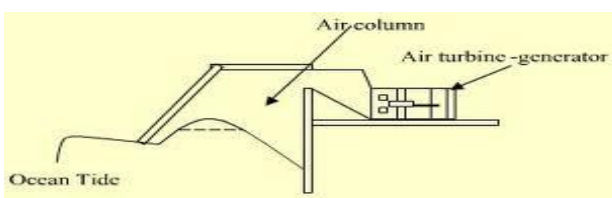


Fig: Tidal Energy – Generator.

2.3.3. Use Alternative (renewable) energy sources:

1.Petro plants: The hydrocarbons present in such plants can be converted into petroleum hydrocarbons. The plants belong to Euphorbiacea, Asclepiadacea, Apocynaceae and Sapotaceae and over 385 species have been screened for hydrocarbon content in this 15 species around promising.
The Indian Institute of Petroleum, Dehra Dun has done excellent work in this area particularly in hydro cracking of the crude products. The products obtained from their latex processed bio crude were gases, naphtha, kerosene, gas oil, heavier, coke.

2. Dendro thermal energy (Energy plantations):
Waste lands are being used for plantation of fast growing shrubs and trees with high calorific value. They in turn provide fuel wood, charcoal, fodder, power.

3.Energy from urban waste: Sewage in cities is used for generating gas and electricity.

4. Bagasse- based plants : Bagesse, a waste of sugar mills can be used to generate energy. It is estimated that sugar mills in India can generate 2,000 MW of surplus electricity during crushing season.

Why alternative (renewable) energy sources are required?
1. The importance of solar energy can be emphasized particularly in view of the fact that fossil fuels and other conventional sources are not free from environmental implications.
2. Energy sources which have least pollution, safety and security snags and are universally available and have the best enhance of large scale utilization in future.
3. Hydro-electric power generation is expected to upset the ecological balance existing on earth.
4. Beside space heating, hydro power plants critically pollute the aquatic and terrestrial biotic.
5. Radio active pollutants released from nuclear power plants are chronically hazardous. The commissioning of boiling water power reactors(BWRS) have resulted in the critical accumulation of large number of long lived radio nuclides in water.
6. The dangerous radio waste cannot be buried in land without the risk of polluting soil and under ground water. Nor the waste can be dumped into the rivers without poisoning aquatic life and human beings as well.
7. The burning of coal, oil, wood, dung cakes and petroleum products have well debated environmental problems. The smoke so produced causes respiratory and digestive problems leading to lungs, stomach and eye diseases.
8. The disposal of fly ash requires large ash ponds and may pose a severe problem considering the limited availability of land. Thus the non-conventional sources of energy are needed.

Objectives:

4. To provide more energy to meet the requirement of increasing population.
5. To reduce environmental pollution and
6. To reduce safety and security risks associated with the use of nuclear energy.

2.3.4. Case Studies:

1. Wind energy in India: India is generating 1200MW electricity using wind energy. The largest wind farm of our country is near Kanyakumari in Tamilnadu, which generates 380MW electricity.

2. Hydrogen – fuel cell car: General motor company of China discovered the experimental cars, which run on electric motors fueled by hydrogen and oxygen. The car produced no emission, only water droplets and water vapor come out of the exhaust pipe.

2.4. LAND RESOURCES

2.4.1. Soil as a Natural resource:

Soil is defined as upper layer of the earth differentiated into various horizons and capable of supporting plant life. Land and its soils are considered as important resources of earth. Land as a resource it is essential for development of agriculture and forestry. Soil is one of the most important ecological factors. Plants depend for their nutrients, water supply, and anchorage upon the soil. Soil is actually formed as a result of long- term process of complex interactions leading to the production of mineral matrix in close association with interstitial organic matter-living as well as dead.

Soil has biological system of living organisms as well as some other components. It is preferred to call it a 'soil complex', which has the following five categories of components.

1. Mineral matter 2. Soil organic matter or humus 3. Soil solution or soil water
4. Soil atmosphere: O_2 and CO_2 content not filled with water.
5. Biological system: bacteria, fungi, algae, protozoa, etc.

Some important functions of soil are as under:

1. It provides mechanical support to the flora.
2. Due to its porosity and water-holding capacity, the soil serves as a reservoir of water and supplies water to the plants (even when the land surface is dry).
3. The ion-exchange capacity of soil ensures the availability and supply of micro- and macro-nutrients for the growth of plants, microbes and animals.
4. Soil also helps in preventing excessive leaching of nutrient ions, while maintaining proper P^H
5. Soil contains a wide variety of bacteria (like nitrifying, nitrogen-fixing, organo trophic, etc.), fungi, protozoans, and many other microbes which help in the

decomposition and mineralization of organic matter and regeneration of nutrients.

2.4.2. Land degradation:

Land degradation refers to deterioration or loss of fertility or productive capacity of the soil. Land resources are very much related to natural disasters like volcanic eruptions, earthquakes etc. but it is due to human activities that soil gets polluted. Land degraded by soil erosion, salination, water-logging, desertification, shifting cultivation, urbanization, landslides and soil pollution.

A. Soil erosion:
The top layer of the soil contains nutrient required by the plants. This fertile top soil is most valuable natural resource, at a depth of 15 – 20 cm. The loss of top soil or disturbance of the soil structure is known as 'soil erosion'.
Types of soil erosion:
Based on the rate of soil loss takes place, there are two main types-

1. Normal or geologic erosion: It occurs under normal, natural conditions by it-self without any interference of man. As natural, it is very slow process, unless there is some major disturbance by a foreign agent.
2. Accelerated soil erosion: This type of removal of soil is very rapid and never keeps place with the soil formation. This is the serious type of loss, generally caused by an interference of an agency like man and other animals.

Agents of soil erosion:
Based on the various 'agents' that bring about soil erosion and the 'form' in which the soil is lost during erosion, there are following types-

a. Water erosion: caused by the action of water, which removes the soil by falling on as 'rain drops', as well as by its 'surface flow' action.
b. Wind erosion: Soil loss by wind is common in dry (arid) regions, where soil is chiefly sandy and the vegetation is very poor or even absent. High velocity winds blow away the soil particles.
c. Landslides or Slip erosion: The hydraulic pressure caused by heavy rains increases the weight of the rocks at cliffs which come under gravitational force and finally slip or fall off. Some times entire hillock may slide down.
d. Stream bank erosion: The river during floods splash their water against the banks and thus cuts through them.
e. Deforestation & Over grazing: Effects of deforestation and over grazing are well known on soil loss. In India annual loss of soil nutrients in this way is of the order of 5.37 million tons of N, P, K valued at about Rs. 700 crores.

B. Salination: Salination refers to increase in the concentration of soluble salts in the soil. Poor drainage of irrigation and flood waters results in accumulation of dissolved salts on the soil surface. In arid and semi-arid areas with poor drainage and high temperatures, water evaporates quickly leaving behind a white crust of salts on the soil surface. The high concentration of salts in soil severely affects the water absorption process of the plants, resulting into poor productivity.

C. Water logging: Water logging may be due to surface flooding or due to high water table. Excessive use of canal irrigation may disturb the water balance and create water logging as a result of seepage or rise in the water table of the area. The productivity of water logged soil is severely affected reduce due to lesser availability of oxygen for the respiration of plants.

D. Desertification: Desertification is a slow process of land degradation that leads to desert formation. I is like a 'skin disease' over planet wherein patches of degraded land, eruption separately, gradually join together. For example, Thar desert (India) was formed by the degradation of thousands of hectares of productive land. It may result either due to a natural phenomenon linked to climatic change or due to abusive use of land. Mismanagement of natural resources including land, certain irreversible changes have triggered the break down of nutrient cycles and micro climatic equilibrium in the soil indicating the ones of desert conditions.
Causes of desertification:
a. Deforestation: The process of denuding and degrading a forested land initiates a desert producing cycle that feeds on itself. Since there is no vegetation to hold back the surface run-off, water drains off quickly before it can soak into the soil to nourish the plants or to replenish the groundwater. This increases soil erosion, loss of fertility and loss of water.
b. Over grazing: This is because the increasing cattle population heavily graze in grasslands or forests and as a result denude the land area. The dry barren land becomes loose and more prone to soil erosion. The top fertile layer is also lost and thus plant growth is badly hampered in such soils.
c. Mining and quarrying: These activities are also responsible for loss of vegetative cover and denudation of extensive land areas leading to desertification.
Amongst the most badly affected areas are the sub Saharan Africa, the Middle East, Western Asia, parts of Central and South America, Australia and the Western half of the US.

E. Shifting cultivation: Shifting (Jhum) cultivation, a very peculiar practice of slash and burn agriculture, prevalent among many tribal communities inhabiting the tropical and sub- tropical regions of Africa, Asia and Islands of Pacific Ocean has also laid large forest tracts bare. This practice has led to complete destruction of forests in many hilly areas of India, North-East and Orissa, and caused soil erosion and other associated problems of land degradation.

F. Urbanization: Human activities are responsible for the land degradation of forests, croplands and grass lands. The productive areas are fast reducing because of urbanization i.e., the developmental activities such as human settlements and industries.

G. Landslides: Human activities such as construction of road and railway, canal, dams and reservoir and mining in hilly areas have affected the stability of hill slopes and damaged the protective vegetation cover above and below roads and other such developmental works. This has upset the balance of nature, making such areas vulnerable to landslides.

H. Soil pollution: Industrial effluents contain high concentration of chemicals. For example Paper and Pulp industrial waste water contain organic matter, Soap industrial waste water contain fatty acids, fertilizer industrial waste water contain chemicals etc. these chemicals degrade the soil fertility and loss of micro organisms.

2.4.3. Man induced landslides:

Landslides are the downward and outward movement of a slope composed of earth materials such as rock, soil, artificial fills. Other names of landslides are rock slide, debris slide, earth flow and soil creep.

During construction of roads and mining activities huge portion of fragile mountainous areas are cut and thrown into adjacent areas and streams. These land masses weaken the already fragile mountain slopes and lead to landslides.

Harmful effect of landslides:

1. Landslides increase the turbidity of nearby streams, thereby reducing their productivity.
1. Destruction of communication links
2. Loss of habitat and biodiversity
3. Loss of infrastructure and economic loss.

Causes of landslides:

1. Removal of vegetation: Deforestation in the sloppy area creates soil erosion, which lead to landslides.
2. Underground mining: These activities cause subsidence of the ground.
3. Transport: Due to the movement of buses and trains in the unstable sloppy region causes landslides.
4. Addition of weight: Addition of extra weight or construction on the slope areas leads to landslides.
5. Ground water level: Over exploitation of ground water also leads to landslides.

2.4.4. Land use / Land cover Mapping:

Land use is the way in which, and the purposes for which, human beings employ the land and its resources: for example, farming, mining, or lumbering. Land cover describes the physical state of the land surface: as in cropland, mountains, or forests. The term land cover originally referred to the kind and state of vegetation (such as forest or grass cover), but it has broadened in subsequent usage to include human structures such as buildings or pavement and other aspects of the natural environment, such as soil type, biodiversity, and surface and groundwater.

Land cover is affected by natural events, including climate variation, flooding, vegetation succession, and fire, all of which can sometimes be affected in character and magnitude by human activities. Both globally and in the India, though, land cover today is altered principally by human use: by agriculture and livestock raising, forest harvesting and management, and construction. There are also incidental impacts from other human activities such as forests damaged by acid rain, from fossil fuel combustion and crops near cities damaged by troposphere ozone resulting from automobile exhaust.

Information on the spatial spread and monitoring the dynamics of the land use/land cover is the basic prerequisite for planning and implementing various developmental activities. Apart from this, nation wide land use information becomes important from the point of view of addressing changing pattern in land use/land cover and also fan overall reporting on the nation’s land use/land cover scenario.

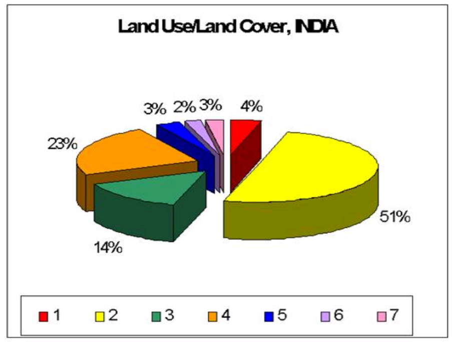
Increasing human interventions and unfavorable bio-climatic environment has led to transformation of large tracts of land into wastelands. Satellite remote sensing plays an important role in generating information about the latest land use-land cover pattern in an area and its temporal changes through times. The information being in digital form can be brought under Geographical Information System (GIS) to provide a suitable platform for data analysis, update and retrieval.

NRSC has carried out various studies on land use/land cover mapping, land cover change, etc. NRSC has carried out land use/land cover mapping for Agro-Climatic Zone Planning covering the 442 districts in the country on 1:250,000 scale using two season (Kharif and Rabi cropping season) data for Planning Commission, Govt. of India.

The main objective of the study was to analyze the land use/cover based on two seasons WiFS data (Kharif Sep-Nov 1988 & Rabi Jan-March 1989) for the whole country and organize/maintain a systematic database on land use/land cover.

Area under broad land use / land cover categories in India (1988 -89)

Category	Area (00 ha.)	%
Builtup Land	139137	4.24
Agricultural Land	1652443	50.26
Forest	476226	14.49
Wastelands	755153	22.98
Waterbodies	106108	3.22
Others	74899	2.28
Unsurveyed	83294	2.53
TOTAL	32872630	100.00



2.5 FOREST RESOURCES

A forest is a biotic community which is predominantly composed of trees, shrubs or any woody vegetation and often with a closed canopy (over-hanging covering). Approximately one third of the earth's total land area is covered by forests. Forests and wildlife are essential for ecological balance of an area. Forests are important components of our environment and economy. Forest ecosystem is dominated by trees, their species content varying in different parts of the world.

2.5.1 Forests have following three types of functions:

1) Productive functions: The productive functions of forests include the production of Timber- for furniture items, Wood- which is used as fuel, raw material for various industries as paper and pulp Industries, news-print, board and packing articles (for fruit, tea etc.), matches, sports goods etc. Indian forests also supply minor forest products- like canes, gums, resins, dyes, tannins, lac, fibers, flocks, medicines, Kath etc. Tribal people forests also provide food (tuber, roots, leaves, fruits, meat from birds and other animals) and Medicare.

2) Protective functions: These functions include conservation of soil and water, prevention of drought and protection against wind, cold, radiation, noise, sights and smells, etc.

3) Regulative functions: Forests are a major factor of environmental concern, providing protection to wildlife, help in balancing the gaseous (CO_2, O_2) cycles of atmosphere, tend to increase local rainfall and water holding capacity of soil, maintain the soil fertility, regulate the earth's temperature, water cycles, check soil erosion, land slides, shifting of sand and silting and reduce flood havoc.

2.5.2 Over- exploitation:

Human beings have depended heavily on forests for food, medicine, shelter, wood and fuel. With growing, civilization the demands for raw material like timber, pulp, minerals, fuel wood etc. more resulting in large scale logging, mining, road-building and clearing of forests. Excessive use of fuel wood and charcoal, expansion of urban, agricultural and industrial areas and overgrazing have together led to over-exploitation of our forests leading to their rapid degradation.

2.5.3 Deforestation (forest destruction):

Total forest area of the world in 1900 was nearly 7000 M.ha. By 1975 it was reduced to 2890 M.ha. By 2005 it was reduced to 2370 M.ha. and if continues like this by 2015 it would be merely 2000 M.ha.

Destruction of biotic potential of land leads to desertification. Such problem arises due to over grazing, indiscriminate felling of trees and over exploitation of land resources.

The disadvantaging effects of deforestation in India include soil, water and wind erosions, estimated to cost over 16,400 crores every year.

Deforestation has a major impact on the productivity of our croplands. This happens in two ways- 1. Soil erosion increases manifold and the soil actually gets washed, leading to an accentuated cycle of floods and drought. 2. But equally important is the impact of the shortage of firewood on the productivity of our croplands.

<u>Causes of deforestation:</u>

- Forests destroyed for growing demand of the cities.
- Local cattle, goats, sheep etc. destroy the vegetation.
- Shivalik sal forests were over exploited for industrial use (railway sleepers etc.)
- Demand of fuel wood and timber due to increase population.
- Construction of roads, railways, houses, dams and power stations.
- The faulty planning of Govt. and its negligence towards forests.
- The green house effect.

<u>Effects of deforestation:</u>

Deforestation affects human life in several ways. The main effects of deforestation are as follows:

1. Forests, particularly on mountains, provide considerable protection from floods by trapping and absorbing precipitation (i.e rain, snow etc) and slowly releasing it later. When the forest is removes, the amount of runoff water flowing into rivers and streams increases several fold causing frequent floods.

2. Deforestation results in increased soil erosion and decreased soil fertility. In drier areas, deforestation can lead to the formation of deserts.

- on an average India is losing about 6,000 million tons of top soil annually due to water erosion in the absence of trees.

3. Deforestation causes the extinction of forests dwelling plant, animal and microbial species resulting into a loss if irreplaceable genetic resources.

4. Deforestation also threats indigenous (tribal) people whose culture and physical survival depend upon the forests.

5. Deforestation induces regional and global climate change. The climate has become warmer due to lack of humidity in the deforested regions.
6. The pattern of rainfall has changed in deforested areas. The rainfall has declined and droughts have become common.
7. Deforestation also contributes to global warming stored carbon into the atmosphere as CO_2, which is a greenhouse gas.
8. Human beings are deprived of benefits of forest trees and wild animals.

2.5.4 Timber and Wood:

Total world wood consumption is about 360 million tons i.e. about 360 million m3/ year, more than steel and plastics together, weight- wise commercial timber and round wood (unprocessed logs) are used to make lumbers, plywood, veneer, particle board and chip board. Approximately 15 crore people depend only on fuel wood for their energy. Average quantity of wood requirement in the world is 1 m3/year/person for cooking and heating.
*China suffered a lot from erosion and floods as it has cut millions of trees 1000 years ago. However, 4.5 million hectares/ year were planted between 1983 – 1993.

Effects of timber extraction:

There has been unlimited exploitation of timber for commercial use. Commercial/industrial demand could out-strip supply leading to decimation of forests, particularly the wood.
The major effects of timber extraction on forests and tribal people include:
* poor logging results in a degraded forests.
* soil erosion, especially on slopes.
* sedimentation of irrigation systems.
* floods may be intensified by cutting of trees on upstream watersheds.
* loss of biodiversity.
* climatic changes, such as lower precipitation.
* new logging roads permit shifting cultivators and fuel wood gatherers to gain access to logged areas and fell the remaining trees.
* loss of non-timber products and loss of long-term forest productivity on the site affect the subsistence economy of the forest dwellers.
* forest fragmentation, the reduction of a large block of forest to many smaller tracts, promotes loss of biodiversity because some species of plants and animals require large continuous areas of similar habitat to survive.
* species of plants and animals, which may occupy marrow ecological niches and whose potential value to humans is unknown, may be eliminated.
* indigenous people may be forced into a new way of life for which they are unprepared.
* exploitation of tribal people by the contractors.
* cutting of more trees than permitted in a particular area by the greedy contractors.

2.6. Citizen’s role in conservation of Natural resources:

You and I share equal responsibility in keeping our environment clean and satisfy our quest for green earth, clean air and pure water.

1. Activities should be organized to promote the awareness of the levels of pollution and their effects and impacts.
2. The right to information, that concerns their own welfare, should be asserted.
3. The provisions of public interest litigation to seek legal redress case of victimization by pollution or violation of norms should be encouraged.
4. Citizens should be –

Co-ordination among the involved Governmental departments to protect the environment.

How can individuals help in conservation of different resources:

Conserve water:

- Don’t keep water taps running while brushing, shaving, washing or bathing.
- In washing machines fill the machine only to the level required for your clothes.
- Install water-saving toilets that use not more than 6 lit. per flush.
- Check for water leaks in pipes and toilets and repair them promptly. A small pin-hole sized leak will lead to the wastage of 640 lit. of water in a month.
- Reuse the soapy water of washings from clothes for washing off the courtyards, driveways. etc.
- Water the plants in your kitchen-garden and the lawns in the evening when evaporation losses are minimum. Never water the plants in mid-day.
- Use drip irrigation and sprinkling irrigation to improve irrigation efficiency and reduce evaporation.
- Install a small system to capture rain water and collect normally wasted used water from sinks, cloth-washers, bath-tubs etc. which can be used for watering the plants.
- Build rain water harvesting system in your house. Even the president of India is doing this.

Conserve Energy:

- Turn off lights, fans and other appliances when not in use.
- Obtain as much heat as possible from natural sources. Dry the clothes in sun instead of drier if it is a sunny day.
- Use solar cooker for cooking your food on sunny days which will be more nutritious and will cut down on your LPG expenses.
- Build your house with provision for sunspace which will keep your house warmer and will provide more light.
- Grow deciduous trees and climbers at proper places outside breeze and shade. This will cut off your electricity charges on coolers and air-conditioners. A big tree is estimated to have a cooling effect equivalent to

five air conditioners. The deciduous trees shed their leaves in winter. Therefore they do not put any hindrance to the sunlight and heat.

- Drive less, make fewer trips and use public transportations whenever possible. You can share by joining a car-pool if you regularly have to go to the same place.
- Add more insulation to your house. During winter close the windows at night. During summer close the windows during days if using an A.C. Otherwise loss of heat would be more, consuming more electricity.
- Instead of using the heat convector more often wear adequate woolens.
- Recycle and reuse glass, metals and paper.
- Try riding bicycle or just walk down small distances instead of using your car or scooter.
- Lower the cooling load on an air conditioner by increasing the thermostat setting as 3-5% electricity is saved for every one degree rise in temperature setting.

<u>Protect the soil:</u>

- While constructing your house, don’t uproot the trees as far as possible. Plant the disturbed areas with a fast growing native ground cover.
- Grow different types of ornamental plants, herbs and trees in your garden. Grow grass in the open areas which will bind the soil and prevent its erosion.
- Make compost from your kitchen waste and use it for your kitchen-garden or flower-pots.
- Do not irrigate the plants using a strong flow of water, as it would wash off the soil.
- Better use sprinkling irrigation.
- Use green manure and mulch in the garden and kitchen-garden which will protect the soil.
- If you own agricultural fields, do not over-irrigate your fields without proper drainage to prevent water logging and Stalinization.
- Use mixed cropping so that some specific soil nutrients do not get depleted.

<u>Promote sustainable agriculture:</u>

- Do not waste food. Take as much as you can eat.
- Reduce the use of pesticides.
- Fertilize your crop primarily with organic fertilizers.
- Use drip irrigation to water the crops.
- Eat local and seasonal vegetables. This saves lot of energy on transport, storage and preservation.
- Control pests by a combination of cultivation and biological control methods.

Review Questions:

1. Discuss the various types of natural resources and their associated problems.
2 . Write a brief note on:
   a) Non-renewable resources   b) Renewable resources
   c) Alternative energy sources
3. a) Define mining and discuss the various steps involved in mining.
   b) Write a detailed note on uses and effect of minerals.
4. a) Explain various steps involved in hydrological cycle with flow chart.
   b) Discuss the advantage and problems associated with dams.
5. a)Write a detailed note on drought and Flood.
   b) Write short notes on land as a resource.
6. Describe the main factors of land degradation.
7. Write a note on a) Land use / Land cover Mapping. b) Man induced land slides.
8. Describe the role of individual in conservation of natural resources.
9. Discuss about the problems due to over exploitation of forest resources.

Fill in the blanks

1. **Renewable** resources are inexhaustible resources which can be generated within a given span of time.
2. **Non-renewable** resources can not be generated.
3. Plants use **CO_2** gas for photosynthesis.
4. Deforestation means **cutting** of forests.
5. **33**% of geographical area of a country should be forest area.
6. Maximum number of dams in India are in the **Maharastra** state.
7. **Tehri** dam is the highest dam on river Bhagirathi in Uttaranchal.
8. The **Bhakra** dam on river Satluj in H.P. is the largest dam in terms of capacity.
9. Ecological issue related with Tehri Dam are taken up by Sh. **Sundarlal Bahuguna** the leader of Chipko movement.
10. Environmental activist Medha Patkar has taken up issues related to **Narmada** Dam.
11. About **97** % of the earth's surface is covered by water.
12. Only **3** % of total water on earth is readily available to us in the form of groundwater and fresh water.
13. A layer of sediment or rock that is highly permeable and contains water (ground water) is called an **aquifer.**
14. Aquifers which are overlaid by permeable earth material and are recharged by seeping water are called **unconfined** aquifer.
15. Aquifers which are sandwiched between two impermeable layers of rocks or sediments are called **confined** aquifers.
16. **Drought** conditions are created when annual rainfall is below normal and less than evaporation.
17. The Cauvery river water is a bone of contention between **Karnataka** and **Tamilnadu** states.
18. Uranium mining is done in **Nalgonda** in A.P.

19. **Aluminium** can be extracted from bauxite ore.
20. Excessive use of fertilizers cause **Micronutrient** imbalance in soil.
21. Eutrophication of lakes is caused the excessive presence of **nitrates** and **phosphates.**
22. In water logged soils the plant roots do not get adequate **O_2** for respiration.
23. Ocean tides are produced by gravitational forces of **sun** and **moon**.
24. In India **Gulf cambay, Gulf of Kutch, Sundarban** deltas are the tidal power sites.
25. For operating ocean thermal energy conversion a difference of **20**^oC or more is required between surface and deeper water of ocean.
26. **Euphorbian**, Palms crops are latex containing plants rich in hydrocarbons.
27. Biogas is produced by **anaerobic** degradation (in the absence of oxygen) of Biological wastes.
28. Gashol is a mixture of **ethanol** and **gasoline**.
29. 95% of natural gas is **methane**.
30. Nuclear energy by nuclear fission is generated when certain isotopes are bombared by **neutrans**.
31. Terrace farming is practiced as a soil conservation measure in **high rainfall** areas.
32. Inadequate drainage and poor quality irrigation water often lead to **water logging** of soils.

Tick the right answer

1. During photosynthesis trees produce [a]
   a) Oxygen b) Carbon dioxide c) Nitrogen d) Carbon monoxide
2. Forests prevent soil erosion by binding soil particles in their [c]
   a) Stems b) Leaves c) Roots d) Buds
3. Wood pulp is used for making [c]
   a) Lumbar b) Chipboard c) Paper d) Plywood
4. Deforestation rate is alarming in [b]
   a) Temperate countries b) Tropical countries c) Polar region d) None
5. Major causes of deforestation are [d]
   a) Shifting cultivation b) Fuel requirements
   c) Raw materials for industries d) All of these
6. Major consequences of deforestation are [d]
   a) Destruction of natural habitat of wild species
   b) Disturbances in hydrological cycle
   c) Soil erosion d) All of these
7. Per capita use of water is the highest in [a]
   a) USA b) India c) Kuwait d) Indonesia
8. Ground subsidence occurs due to
   a) Withdrawal of more groundwater than its recharge
   b) More recharge of groundwater than its withdrawal
   c) Equal rates of recharge and withdrawal
   d) None of the above.

9. Which of the following dreams to become the water super power in the middle east countries. [d]
   a) Kuwait b) Syria c) Jordan d) Turkey
10. The Satluj – Yamuna Link (SYL) canal dispute is between [a]
    a) Punjab and Haryana b) Karnataka and Tamilnadu
    c) Delhi and U.P d) All of the above
11. Over grazing results in [b]
    a) Productive soils b) Soil erosion
    b) Retention of useful species d) All of these
12. Blue baby syndrome (methamoglobinemia) is caused by the contamination of water due to [d]
    a) Phosphates b) Sulphur c) Arsenic d) Notrates
13. Accumulation of non-biodegradable materials in the food chain is called [a]
    a) Biomagnification b) Detoxification
    c) None of these d) Both of these
14. Natural geysers which operate due to geothermal energy are present in [d]
    a) Manikaran in Kullu b) Sohana in Haryana
    c) None of them d) Both of these
15. Biomass energy can be obtained from [d]
    a) Energy plantations b) Petro crops
    c) Agricultural and urban waste biomass
    d) All of these
16. Which of the following types of coal has maximum carbon and calorific value?
    a) Anthracite b) Bituminous c) Lignite d) Wood coal [a]
17. Nuclear energy can be generated by [c]
    a) Nuclear fusion b) Nuclear fission c) Both of these d) None of these
18. The minimum time needed for the formation of one inch of top soil is [d]
    a) 10 years b) 50 years c) 100 years d) 200 years
19. Minimum disturbance is caused to SOCI during [b]
    a) Contour farming b) No- till farming
    c) Terrace farming d) Alley cropping
20. Which of the following is responsible for desertification? [d]
    a) Deforestation b) Overgrazing c) Mining d) All of these
21. Removing natural resources from the environment and adding to environmental problems through pollution are major factors in a process called [c]
    a) the greenhouse effect b) the biotechnological revolution
    c) environmental degradation d) the green revolution
22. Approximately 70 percent of the earth is covered by water. Of this amount, approximately _____ percent of it is suitable for human use. [a]
    a) less than one b) 25 c) 10 d) 50

23. The largest amount of usable water found on earth is used for ----------- [a]
    a) crop irrigation b) recreation c) household use d) industrial uses

Write whether the following statements are true or false

1. Surface water is more in quality than the groundwater. **False**
2. Networking of rivers is being proposed at national level to deal with the problem of Floods. **True**
3. Under Indus Water Treaty, Indus, Jhelum and Chenab were allocated to India and Satluj, Ravi and Beas to Pakisthan **True**
4. Small dams are environmentally more sustainable than big dams. True
5. Copper mines are located in Khetri (Rajasthan). **True**
6. Mining of Uranium exposes local people to radioactive hazards. **True**
7. Dichlorodiphenyl trichloroethane is the technical name of aldrin. **False**
8. The major source of salinization of soil is excessive irrigation. **True**
9. Solar heat can not be used to operate street lights, water pumps, televisions, calculators etc. **True**
10. Solar cells are made up of thin wafers of semi-conductor materials like silicon and gallium. **True**
11. Ideal location for installation of wind-mills are coastal regions, open grasslands, hilly regions. **True**
12. Hydropower also causes environmental pollution. **False**
13. Geothermal energy is produced as a result of fission of radioactive material naturally present in rocks. **True**
14. Burning of dung produces biomass energy but doesn't destroy essential nutrients like N and P. **False**
15. Sludge left over in the biogas plant cannot be used as a fertilizer. **False**
16. Ethyl mercaptane (a foul smelling gas) is added to odor less LPG for instantaneous detection of any leakage. **True**
17. Soil erosion helps to maintain soil fertility. **False**
18. Alley cropping is intercropping of crops with trees or shrubs. **True**
19. Gully erosion is a mild type of soil erosion found in regions of low rainfall. **False**
20. The soil below a depth of 20 cm is the fertile soil. **False**